

# fk\_index\_patch – guide

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**Revision: 1 Last modified:** 2026-09-26T12:58:00Z

Source: `constitution/scripts/codegraph/fk_index_patch.py` (259 lines), plus the patch registry `runner_patches/__init__.py` (35 lines) and the shared contract `runner_patches/common.py` (59 lines). Citations below are `fk_index_patch.py:<line>` unless another file is named.

Per-patch guides (one per registered patch): [fkidx1](#), [lockfix1](#), [datafrag1](#), [resolve1](#), [resolve2](#), [mcpro1](#).

## Overview

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`fk_index_patch.py` builds a **private, version-keyed, PATCHED copy** of the CodeGraph platform package ( `@colbymchenry/codegraph-linux-x64` , the bundle that holds `bin/codegraph` , `node` and `lib/dist/*.js` ). The shared global install is never modified: the tool copies it ( `cp -a --reflink=auto` , line 225), rewrites a handful of JavaScript files in the copy, and writes a `RUNNER_RECEIPT.json` that records exactly what was changed (lines 3-24, 62-63, 225-242). The private copy is disposable and regenerable: delete the directory and re-run (lines 10-12).

It exists because stock CodeGraph 1.6.0 has performance and safety defects that matter on a 580K-file tree (FK-cascade full scans during bulk indexing, a lock-stealing 2-minute age override, parse-pool timeouts on data-fragment headers, an  $O(\#refs \times \#files)$  resolve query, a heap OOM in the Swift->ObjC bridge, and an MCP server that writes to the index). Each defect is fixed by one small **runner patch** module; this tool selects patches, applies them fail-closed, and packages the result.

The patch set is **selected** by `--patches` (or the registry default), and every selected patch is **applied** against a shared in-memory view of the package so a later patch sees the edits of earlier ones ( `runner_patches/__init__.py:5-8` ). Nothing is written to disk until every selected patch has planned successfully (lines 126-146, 201, 225-228).

## Prerequisites

- `python3` , standard library only (imports lines 71-78).
- GNU `cp` with `-a` and `--reflink=auto` (line 225).
- `npm` on `PATH` , **only** when `--src` is not given: the default source is resolved with `npm root -g` (timeout 60 s, lines 104-107).
- Linux `/proc` for the "runner in use" guard (lines 149-163). Where `/proc` is absent the guard silently finds no users (line 153: the loop iterates over nothing), so the protection is off, not an error.
- The source package must be **pristine** (unpatched). Every patch except `fkidx1` refuses an already-patched tree with a message saying so (see the per-patch guides); `fkidx1` reports `NOT_NEEDED` instead (verified, below).
- It writes under `~/.cache/helix/codegraph_runner/<key>` by default (lines 207-209). **When experimenting, always pass an explicit `--dst` under a scratch directory**: the default directory is the sanctioned runner cache that live writers and MCP servers execute from. The `--force` path refuses to replace a directory whose path appears in any live process command line (lines 217-220), but do not rely on that as the only safeguard.

## Usage

```
fk_index_patch.py [--src <platform package dir>] [--dst <runner dir>]
                  [--patches id,id,...] [--force] [--print-bin] [--list-patches]
```

| Flag                                  | Default                                                                                                                | Meaning                                                                                                                                                                                                      | Source                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <code>--src</code><br>DIR             | the installed platform package                                                                                         | Platform package to patch (read-only). Default: first child of <code>&lt;npm root -g&gt;/@colbymchenry/codegraph/node_modules/@colbymchenry/</code> (sorted) that contains <code>lib/dist/db/index.js</code> | lines 182, 104-116, 199 |
| <code>--dst</code><br>DIR             | <code>\$</code><br><code>{HELIX_CODEGRAPH_RUNNER_DIR: \$HOME/.cache/helix/code-graph_runner}/&lt;runner key&gt;</code> | Where to build the runner. An empty <code>HELIX_CODEGRAPH_RUNNER_DIR</code> counts as unset ( or , line 207)                                                                                                 | lines 183, 207-209      |
| <code>--patches</code><br>a, b, c     | every registered patch except opt-in ones (see below)                                                                  | Comma list; whitespace trimmed, empty items dropped; if nothing is left:<br><code>--patches given but empty</code> , exit 2                                                                                  | lines 184, 194-198      |
| <code>--force</code>                  | off                                                                                                                    | Rebuild even when a valid receipt exists (skips the reuse check)                                                                                                                                             | lines 185, 212          |
| <code>--print-bin</code>              | off                                                                                                                    | On success print only the path of <code>bin/codegraph</code> instead of the JSON report                                                                                                                      | lines 186, 215, 245     |
| <code>--list-patches</code>           | off                                                                                                                    | Print <code>&lt;id&gt;\t&lt;summary&gt;</code> for the <b>default</b> patch set and exit 0; needs no <code>--src</code>                                                                                      | lines 187, 190-193      |
| <code>-h</code> , <code>--help</code> |                                                                                                                        | argparse help                                                                                                                                                                                                | line 181                |

Runner key = `<codegraph version>-<applied patch ids joined by '- '>`  
(line 206); ids of patches that reported NOT\_NEEDED are left out. Example on  
stock 1.6.0 with the default set: `1.6.0-fkidx1-lockfix1-datafrag1-resolve1-  
resolve2` .

## Examples

1. **List the default patch set** — VERIFIED (ran: `fk_index_patch.py --list-patches`; saw: five tab-separated rows `fkidx1`, `lockfix1`, `datafrag1`, `resolve1`, `resolve2`, `exit 0`) . `mcpro1` is **not** listed even though `--help` names it, because `--list-patches` uses the default set and `mcpro1` is opt-in ( `runner_patches/__init__.py:18,24-25` ).
2. **Build a runner in a scratch directory** (never the real cache) — VERIFIED (ran: `fk_index_patch.py --src <tiny fake package> --dst <scratch>/run --patches fkidx1`; saw: JSON with "status": "CREATED", runner\_key "9.9.9-fkidx1", receipt\_schema 2, `exit 0`) . The fake package is the fixture builder `make_pkg` from `tests/test_patch_machinery.py` ; a real package copy is ~282 MB, so the scratch build used the tiny fixture. Command shape for a real package ( NOT RUN here: it would copy 282 MB; the test suites in the Last verified section exercise that path for `resolve1/resolve2/mcpro1`):

```
python3 scripts/codegraph/fk_index_patch.py \  
  --src "$(npm root -g)/@colbymchenry/codegraph/node_modules/@colbymchenry/codegraph-  
  linux-x64" \  
  --dst /scratch/runner_test --patches resolve1
```

3. **Get only the launcher path** — VERIFIED (ran: same command with `--print-bin` on an existing valid runner; saw: the single line `<dst>/bin/codegraph`, `exit 0`) . This is how the callers use it:  
`codegraph_safe.sh` runs it with `--print-bin`  
( `codegraph_safe.sh:455-456` ) and `codegraph_mcp.sh` with `--patches mcpro1 --print-bin` ( `codegraph_mcp.sh:58` ).
4. **Second run is a no-op** — VERIFIED (ran: same build command twice; saw: "status": "REUSED" on the second run, `exit 0`) .

5. **Dry-run planning against the real package without copying anything** — the CLI has no dry-run flag, but `plan_all` is importable and read-only. **VERIFIED** (ran: `python3` with `sys.path` at `scripts/codegraph`, `import fk_index_patch; fk_index_patch.plan_all(<stock 1.6.0 dir>, None)`; saw: `applied = ['fkidx1', 'lockfix1', 'datafrag1', 'resolve1', 'resolve2']`, edited files = `db/index.js`, `extraction/index.js`, `resolution/frameworks/swift-objc.js`, `resolution/frameworks/vue.js`, `resolution/index.js`, `utils.js`; and each of the six patches, planned alone, reports **APPLIED** against `stock 1.6.0`) . Importing the module only sets `sys.dont_write_bytecode` and edits `sys.path` (lines 80-82); it builds nothing.
6. **Refusals leave nothing behind** — **VERIFIED** (ran: `--patches fkidx1,mcpro1 / --patches nope / --patches ""`; saw: `exit 2` each with a one-line JSON `{"status": "REFUSED", "reason": ...}` on `stderr` and no `<dst>` directory created) .
7. **NOT\_NEEDED** — **VERIFIED** (ran: `--patches fkidx1` against a fake package whose `queries.js` no longer contains `"INSERT OR REPLACE INTO nodes"`; saw: JSON `{"status": "NOT_NEEDED", "reason": "fkidx1: queries.js no longer uses INSERT OR REPLACE INTO nodes", ...}` on `stdout`, `exit 3`, no `<dst>` created) .

## Outputs

**stdout** on success (exit 0): with `--print-bin` , one line (the launcher path); otherwise a JSON object `{"status": "CREATED"|"REUSED", "runner": <bin path>, ...receipt fields}` (lines 215, 245). On exit 3, a JSON `{"status": "NOT_NEEDED", "reason": ..., "patches": {...}}` is printed **even with** `--print-bin` (lines 202-205): callers must test the exit status before treating `stdout` as a path. **stderr** on exit 2: one JSON line `{"status": "REFUSED", "reason": ...}` (lines 250-255).

**Files:** `<dst>/` (patched copy of the package) and `<dst>/RUNNER_RECEIPT.json` (lines 229-239):

| Receipt field                                                         | Meaning                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>receipt_schema</code>                                           | <code>2</code> ( <code>RECEIPT_SCHEMA</code> , line 88)                                                                                                                                                                                                                                                                                          |
| <code>runner_key</code>                                               | <code>&lt;version&gt;-&lt;applied ids&gt;</code> (line 206)                                                                                                                                                                                                                                                                                      |
| <code>patch_set</code>                                                | list of applied patch ids in registry order                                                                                                                                                                                                                                                                                                      |
| <code>codegraph_version</code>                                        | version from the source <code>package.json</code> (lines 119-123)                                                                                                                                                                                                                                                                                |
| <code>source</code>                                                   | absolute <code>--src</code> path                                                                                                                                                                                                                                                                                                                 |
| <code>source_dbjs_sha256</code> ,<br><code>patched_dbjs_sha256</code> | sha256 of <code>lib/dist/db/index.js</code> before / after<br>( <code>DBJS_REL</code> , line 89)                                                                                                                                                                                                                                                 |
| <code>patches</code>                                                  | per patch id: <code>status</code> ( <code>APPLIED</code> or <code>NOT_NEEDED</code> ),<br><code>summary</code> , and for <code>APPLIED</code> <code>anchors</code> ,<br><code>files{rel: {source_sha256,</code><br><code>patched_sha256}}</code> , <code>detail</code> ; for <code>NOT_NEEDED</code><br><code>reason</code> (lines 134, 143-144) |
| <code>created_utc</code>                                              | UTC timestamp <code>%Y-%m-%dT%H:%M:%SZ</code>                                                                                                                                                                                                                                                                                                    |
| legacy fields                                                         | only when <code>fkidx1</code> is applied: <code>patch_id: "fkidx1"</code><br>plus <code>fkidx1</code> 's <code>kept_indexes</code> , <code>lists_before</code> ,<br><code>lists_after</code> (lines 235-236) — kept for con-<br>sumers of the pre-registry single-patch tool                                                                     |

## Exit codes

| Exit | Meaning                                                                                                                                                                                                                                                                                                                                                             | Source                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 0    | Runner ready: CREATED now, REUSED, or <code>--list-patches</code> printed                                                                                                                                                                                                                                                                                           | lines 190-193, 215, 245-246                                                                                                  |
| 2    | Fail-closed: an anchor did not occur exactly once, unknown/empty/exclusive-conflicting <code>--patches</code> , a planned edit that would be a no-op, unreadable source, a runner directory in use by a live process, a launcher that is not executable, an <code>OSError</code> , a failed <code>cp</code> , or a corrupt receipt ( <code>JSONDecodeError</code> ) | lines 138-140, 198, 217-220, 243-244, 250-255; <code>com-mon.py:52-59</code> ; <code>runner_patches/__init__.py:26-34</code> |
| 3    | Every selected patch reported NOT_NEEDED (the hazard is provably absent upstream)                                                                                                                                                                                                                                                                                   | lines 202-205                                                                                                                |

Argparse errors use argparse's own exit 2 (usage) as well.

## How the patch set is selected and applied

### Selection ( `runner_patches/__init__.py` )

- `REGISTRY = ["fkidx1", "lockfix1", "datafrag1", "resolve1", "resolve2", "mcpro1"]` (line 17). Order matters: patches are planned and applied in registry order on a shared view, whatever order `--patches` lists them in (line 30). Verified: `--patches resolve2,fkidx1 loads fkidx1, resolve2`.
- **Default** (no `--patches`, `load(None)`): every registry id that is **not** opt-in (lines 24-25) — today the five writer patches. The default runner key on stock 1.6.0 is therefore `1.6.0-fkidx1-lockfix1-datafrag1-resolve1-resolve2` ( `VERIFIED`, from `plan_all`, see Example 5).

- `OPT_IN = {"mcpro1"}` (line 18): selectable with `--patches mcpro1` but never part of the default set, so the read-only MCP runner cannot end up in a writer runner by omission (lines 9-13).
- `EXCLUSIVE = {"mcpro1"}` (line 19): a patch that defines a different *kind* of runner and may not be combined with any other patch; `--patches mcpro1,fkidx1` is refused with `patch 'mcpro1' is exclusive ... cannot be combined with: fkidx1` (lines 31-34; VERIFIED, Example 6). Consequently the runner key `<ver>-mcpro1` identifies the read-only runner exactly.
- Unknown ids: `unknown patch id(s): ... (known: ...)` (lines 26-29).

Note the difference between this tool's default and `codegraph_safe.sh`'s default: the wrapper passes `--patches fkidx1,resolve1` unless overridden ( `codegraph_safe.sh:244` ; it omits `--patches` <sup>1 2 6</sup> <sup>1 4 6</sup> only for `all`, lines 455-456), so the runner key it uses is `<version>-fkidx1-resolve1`, not the five-patch key.

## Application ( `plan_all`, lines `–` ; `common.py` )

1. A `FileView` over `--src` is created ( `common.py:27-42` ): `read(rel)` returns a pending edit if one exists, otherwise the file text.
2. For each selected module, in registry order, `mod.plan(view)` returns a `PatchPlan` ( `common.py:45-59` ) or raises `NotNeeded` / `Refuse` .
3. Every text replacement goes through `PatchPlan.replace_once`, which counts the anchor text and raises `Refuse` unless it occurs **exactly once** ( `common.py:52-59` ). VERIFIED for the five anchor-based patches: for every anchor of `lockfix1`, `datafrag1`, `resolve1`, `resolve2` and `mcpro1`, removing it (0 occurrences) or duplicating it (2 occurrences) in a copy of the real 1.6.0 files produced `Refuse: <id>: anchor '<label>' occurs 0x|2x in <file>` (expected exactly 1) – upstream shape changed, re-derive the patch .
4. `NotNeeded` records `status: NOT_NEEDED` with the reason, skips the patch and leaves its id out of the key (lines 131-135).
5. If a plan's new text equals the old text the tool refuses ( `planned edit of <rel>` is a no-op – refusing, lines 137-140): a patch that silently changes nothing would be a bluff.

6. Source and patched sha256 of every edited file go into the receipt, and the new text is layered into the view for later patches (lines 141-142).
7. Only after all plans succeed is anything copied (lines 221-228).

## The six patches and how they relate

| Id         | Kind                        | Files edited (real 1.6.0)                                                             | Purpose                                                                                                                                                   | De-fault?                        |
|------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| fkidx1     | writer / bulk index         | lib/dist/db/index.js                                                                  | Keep the FK child-key indexes through the bulk-index windows so INSERT OR REPLACE INTO nodes under foreign_keys=ON never degrades into cascade full scans | yes                              |
| lock-fix1  | writer / safety             | lib/dist/Utils.js                                                                     | A codegraph.lock is stale only when its PID is not a live codegraph process; removes the 2-minute age override                                            | yes                              |
| data-frag1 | writer / bulk index         | lib/dist/extraction/index.js                                                          | Store raw number-fragment C headers and NUL-containing files as skip rows instead of timing out the parse pool                                            | yes                              |
| re-solve1  | writer / resolve            | lib/dist/resolution/index.js , .../frameworks/vue.js                                  | Memoize the resolution context's file-path list per cache generation                                                                                      | yes                              |
| re-solve2  | writer / resolve            | lib/dist/resolution/frameworks/swift-objc.js                                          | Stream (never materialize) the method node set in the Swift->ObjC bridge                                                                                  | yes                              |
| mcpro1     | <b>read-only MCP runner</b> | db/index.js ,<br>Utils.js ,<br>mcp/engine.js ,<br>mcp/index.js ,<br>bin/code-graph.js | Make the runner incapable of mutating a project; for serve --mcp only                                                                                     | <b>no</b><br>(opt-in, exclusive) |

Relations, all verified from `plan_all` on stock 1.6.0:

- The five default patches edit **disjoint files** (six files in total), so today their relative order does not change any result; the shared view and the registry order exist so a future patch can build on an earlier one.
- `fkidx1` and `mcpro1` both edit `db/index.js`, and `lockfix1` and `mcpro1` both edit `utils.js`; that overlap is a fact of the code. The reason stated in the source for exclusivity is that `mcpro1` defines a different runner kind (`runner_patches/__init__.py:10-13`).
- `resolve2` only *reads* `resolution/index.js` (it asserts the context exposes `iterateNodesByKind`, `resolve2.py:36,55-57`); `resolve1` is the patch that edits that file.
- `fkidx1` has an external oracle, `fk_cascade_probe.py` (header line 66 and `fkidx1.py:6-7`), a separate tool that `codegraph_safe.sh` runs to prove stock safe when this tool exits 3 (NOT\_NEEDED) (`codegraph_safe.sh:469-473`).

## Reuse, rebuild and replacement (lines – )

1. Key and destination are computed (206-210).
2. If `<dst>/RUNNER_RECEIPT.json` exists and `--force` is not given, the receipt is read and `receipt_valid` decides reuse (212-216):
  - schema-2 receipt: reuse iff `runner_key` and `patch_set` equal this invocation's, and **every edited file** in `<dst>` hashes to the freshly planned patched text (166-177);
  - legacy receipt (no `patches` key): reuse iff the applied set is exactly `["fkidx1"]`, `patch_id` is `fkidx1`, and both `db/index.js` hashes match (168-171).
3. Otherwise, if `<dst>` exists and any live process command line contains `<dst>/`, the tool refuses (149-163, 217-220). PIDs 1 and the tool's own PID are skipped.

4. `<dst>.partial` is removed if present, the parent directory created, the package copied, the edited files rewritten, the receipt written, the old `<dst>` deleted, and `<dst>.partial` renamed into place (221-242).
5. Last, the launcher `<dst>/bin/codegraph` must be executable, else `Refuse` (243-244). This check runs **after** the rename, so on that refusal the directory has already been swapped in.

## Edge cases

- **Already-patched source.** `VERIFIED` (planned each patch against the output of its own edit) : `lockfix1`, `datafrag1`, `resolve1`, `resolve2` and `mcpro1` raise  
`Refuse: <id>: source is ALREADY patched --src must be the pristine package ; fkidx1 raises NotNeeded: every FK child column is already served by a never-dropped index (so a second pass reports NOT_NEEDED, not an error).`
- **Upstream upgrade.** After a `CodeGraph` upgrade that rewrites the anchored code, the affected patch exits 2 naming the patch id and anchor, and **no runner is produced** (lines 26-32). `fkidx1` is regex-based rather than anchor-based; its shape failures read e.g. `BULK_EDGE_INDEX_NAMES not found/ empty -- upstream shape changed ( VERIFIED , list renamed in a copy).`
- **NOT\_NEEDED is per patch.** A patch that proves its hazard absent is skipped and dropped from the key: `fkidx1` does so when `queries.js` no longer uses `INSERT OR REPLACE INTO nodes`, when `foreign_keys=ON` is gone, or when `schema.sql` has no `REFERENCES nodes(id)` ( `VERIFIED`, each variant built in a copy of the real 1.6.0 files); `lockfix1` does so when `STALE_TIMEOUT_MS` is gone from `utils.js` ( `lockfix1.py:61-65` ); `datafrag1` when the upstream already records `data_fragment_skipped` and `binary_content_skipped` ( `datafrag1.py:110-111` ). `resolve1`, `resolve2` and `mcpro1` have no `NOT_NEEDED` path in their source. Exit 3 only when **all** selected patches are `NOT_NEEDED` (lines 202-205); `codegraph_safe.sh` reacts to exit 3 by requiring `fk_cascade_probe.py` to prove stock safe ( `codegraph_safe.sh:469-473` ).

- **Concurrent use.** Nothing serialises two invocations building the same `<dst>` : both would use the same `<dst>.partial` path (line 221). UNCONFIRMED: behaviour of two simultaneous builds was not run; callers such as `codegraph_safe.sh` are expected to serialise.
- **Disk.** A build copies the whole package, ~282 MB for stock 1.6.0 ( `du -sh` of the installed package). `--reflink=auto` avoids the byte copy only where the filesystem supports reflinks; otherwise it is a full copy.
- **Corrupt receipt.** VERIFIED (ran: build over a `<dst>` whose `RUNNER_RECEIPT.json` contains `{"corrupt"; saw: exit 2 and {"status": "REFUSED", "reason": "JSONDecodeError: Expecting property name enclosed in double quotes: line 1 column 2 (char 1)"}}`); `--force` rebuilds it (lines 212-213).
- **Bad flag.** An unrecognised option is rejected by `argparse` with exit 2 ( `VERIFIED` , `--bogus` ).

## Known discrepancies and gaps (observed – – )

These are facts about the tree on the verification date, recorded so nobody relies on the tool doing more than it does.

1. **tests/test\_patch\_machinery.py** is RED against this tool. Its own header says it reproduces four review findings (P1-P4) against a tiny fake package. `VERIFIED` (ran: `PYTHONDONTWRITEBYTECODE=1 python3 tests/test_patch_machinery.py`; saw: SUMMARY fails=9 P1b P1c P2c P2d P2e P2f P3 P4a P4b; PASS only P1a P2a P2b P4c) . What each failing case expects that the tool does not do today:
  - **P1b/P1c** — a commented-out duplicate of, or a duplicated declaration of, `static BULK_PARSE_INDEX_NAMES = [...]` should refuse with exit 2 and build nothing; today the tool exits 0 and builds a runner (fkidx1's list parser does not assert list-anchor uniqueness).
  - **P2c/P2e** — the reuse check should verify the whole runner tree; today it verifies only the edited files, so a truncated non-edited file or a planted extra file still yields `REUSED` (lines 172-176).

- **P2d/P2f** — after the launcher is deleted, reuse should not print a path to a missing file; today `--print-bin` prints the path of a non-existent launcher (lines 212-216 never test existence).
- **P3** — expects the default patch set to be exactly `fkidx1, resolve1, resolve2` ; the registry default is five patches `( fkidx1, lockfix1, datafrag1, resolve1, resolve2 )`.
- **P4a** — a stale `<dst>.partial` should be removed on the REUSED path; it is only removed on the rebuild path (lines 221-223).
- **P4b** — a corrupt receipt exits 2 as expected, but the error message (a `JSONDecodeError` ) does not mention `--force` . The recovery itself works (P4c PASS: `--force` skips the receipt read, line 212).

UNKNOWN: whether the test is ahead of the tool (RED written first, fix pending) or the tool regressed; the constitution history in this checkout is a single squashed commit for these files, so it cannot be determined here. Reported to the conductor.

- In-source header is stale in four places.** (a) It lists three patches (fkidx1, lockfix1, datafrag1, lines 12-23) but the registry has six; (b) it says `--patches` defaults to "all" (line 38) whereas mcpro1 is opt-in and excluded by default; (c) its example key `1.6.0-fkidx1-lockfix1-datafrag1` (line 43) is not the default key today ( `1.6.0-fkidx1-lockfix1-datafrag1-resolve1-resolve2` ); (d) it cites `tests/run_runner_patch_tests.sh` "(lockfix1/datafrag1 RED/GREEN + mutations)" (line 67): **that file does not exist** ( `git ls-files` finds no such path; control: the same query finds `tests/run_scope_tests.sh` ). This guide follows the code, not the header.
- lockfix1 and datafrag1 have no behavioural test in the tree.** No file under `scripts/codegraph/tests/` contains their runtime markers `( __helixLockOwnerIsCodegraph , __helixDataFragClassify , data_fragment_skipped , binary_content_skipped , STALE_TIMEOUT_MS ;` control: the same query finds `helixAllFilesMemo` ). `owed_report.py` marks them "tested" because three test files merely *name* the ids (patch-set selection arguments, a registry-string mutant), and `codegraph_safe.sh:63-68` says outright that they have zero tests, which is why the wrapper does not use them by default. The two per-patch guides say what was and was not verified.

4. **Dangling link.** The guides for `codegraph_guard` , `codegraph_safe` and `disk_tripwire` link to `codegraph_runner_patches.md` ; no such file exists in `docs/scripts/` .

## Internal behaviour (function map)

| Function                                          | Lines   | Role                                                                                                                                               |
|---------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sha256</code> ,<br><code>sha256_text</code> | 92-101  | File / text digests for receipts and reuse                                                                                                         |
| <code>default_src</code>                          | 104-116 | Locate the installed platform package via<br><code>npm root -g</code>                                                                              |
| <code>package_version</code>                      | 119-123 | Read <code>package.json</code> <code>version</code>                                                                                                |
| <code>plan_all</code>                             | 126-146 | Plan every selected patch on one <code>FileView</code> ;<br>NOT_NEEDED bookkeeping; no-op refusal                                                  |
| <code>runner_in_use</code>                        | 149-163 | PIDs whose real <code>/proc/&lt;pid&gt;/cmdline</code> contains<br><code>&lt;dst&gt;/</code> (never a bare <code>pgrep</code> , per the docstring) |
| <code>receipt_valid</code>                        | 166-177 | Reuse decision (schema-2 and legacy)                                                                                                               |
| <code>main</code>                                 | 180-255 | CLI, key/dst, reuse, copy, write, swap, exit-code<br>mapping                                                                                       |

Legacy re-exports: `FK_RE`, `IDX_RE`, `LIST_RE`, `TABLE_RE`, `compute_patch`, `derive_keep_set`, `parse_lists`, `patch_text` are re-exported from `fkidx1` (lines 84-85) and `default_src` is called by tests as `m.default_src()` ( `tests/test_mcp_readonly.sh:62` ).

## Tests and mutation evidence

| Test                                                                   | Covers                                                                                                                                                            |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>tests/</code><br><code>test_patch_machinery.py</code>            | The tool's own machinery (anchor uniqueness, reuse validation, default set, <code>.partial</code> /receipt handling); 9 of 13 checks FAIL today (above)           |
| <code>tests/test_resolve1.sh</code> ,<br><code>test_resolve2.sh</code> | Build a single-patch runner through this tool ( <code>--patches resolve1 / resolve2</code> ) and check behaviour                                                  |
| <code>tests/test_mcp_readonly.sh</code>                                | Builds <code>--patches mcpro1</code> ; R1 receipt $\geq 3$ anchors, R8 shape mismatch $\Rightarrow$ exit 2 and no runner directory, R13 registry/opt-in/exclusive |
| <code>tests/*_mutations.sh</code>                                      | Paired mutation harnesses for the above                                                                                                                           |
| <code>tests/fixtures/</code><br><code>stub_patch_tool.sh</code>        | Stand-in used by <code>codegraph_safe.sh</code> tests ( <code>test_unit_safe.sh</code> )                                                                          |

Also naming this tool but not run for this guide (they may build into the real runner cache): `test_codegraph_mcp.sh` , `test_codegraph_mcp_preflight.sh` and their mutation harnesses.

## Related scripts

| Script / guide                                                                                                                                               | Relation                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <a href="#">fkidx1</a> , <a href="#">lockfix1</a> , <a href="#">datafrag1</a> , <a href="#">resolve1</a> , <a href="#">resolve2</a> , <a href="#">mcpro1</a> | The six registered patches                                                                                          |
| <a href="#">codegraph_safe</a>                                                                                                                               | Sanctioned bulk-index entry; derives its runner from this tool ( <code>--patches fkidx1,resolve1</code> by default) |
| <code>codegraph_mcp.sh</code>                                                                                                                                | Builds/verifies the <code>mcpro1</code> runner through this tool ( <code>codegraph_mcp.sh:58</code> )               |
| <code>fk_cascade_probe.py</code>                                                                                                                             | Oracle for <code>fkidx1</code> ; proves stock safe when this tool exits 3                                           |
| <a href="#">owed_report</a>                                                                                                                                  | Reports whether this tool has tests and a guide                                                                     |

## Last verified

2026-09-26 (all runs in scratch directories under the session scratchpad; the real runner cache `~/.cache/helix/codegraph_runner` was not touched — its directory listing was identical before and after):

- `--list-patches` , `--help` , and the CLI scenarios of Examples 2-4, 6, 7 on a tiny fake package: outputs as described above.
- `plan_all` against the installed stock 1.6.0 package: each of the six patches applies alone; the default set applies as five and would produce key `1.6.0-fkidx1-lockfix1-datafrag1-resolve1-resolve2` ; `fkidx1` keeps `idx_edges_target_kind` and `idx_unresolved_from_node` on this version.
- Every patched JavaScript file produced by each patch (and by the default set together) passes `node --check` with the package's bundled node (control: `node --check` on a deliberately broken file exits 1).
- `tests/test_resolve2.sh` 3/3 PASS, `test_resolve1.sh` 7/7 PASS, `test_mcp_readonly.sh` 13/13 PASS; mutation harnesses: resolve2 golden + 5/5 mutants killed, resolve1 golden + 5/5 killed, mcp\_readonly 10/10 killed.

- `tests/test_patch_machinery.py` : 9 FAIL / 4 PASS (see Known discrepancies).